

DEPARTMENT OF APPLIED MATHEMATICS AND COMPUTATIONAL SCIENCE**B.Tech. II YEAR (4YDC)****MA 10501: MATHEMATICS-II**

HOURS PER WEEK			CREDITS			MAXIMUM MARKS				
T	P	Tu	T	P	Tu	THEORY		PRACTICAL		TOTAL MARKS
						CW	END SEM	SW	END SEM	
3	--	1	4	--	--	30	70	--	--	100

PRE –REQUISITES: NIL**COURSE OBJECTIVES**

Enable the students to apply the knowledge of Mathematics in various engineering fields by making them

- able to perform elementary operations on matrix and learn to solve systems of linear equations and their application problems.
- solve first order and first degree ordinary differential equations, higher order linear differential equations and their applications .
- understand the basic concepts of probability distributions and curve fitting .
- provide the basic knowledge of functions of complex variable and their applications .

COURSE OUTCOMES

The students will be able to

- CO#1** understand the various applications of the theory of matrices to a wide variety of problems like system of linear equations, inverse of a matrix , Eigen values and Eigen vectors.
- CO#2** classify differential equations according to certain features and solve first order differential equations .
- CO#3** able to solve higher order linear differential equation with applications to simple problems.
- CO#4** acquire the basic knowledge of probability distributions (Binomial,Poisson and Normal) with their applications and fitting of curves using method of least squares.
- CO#5** understand the concept of various functions of a complex variable with the summation of infinite series as an application

COURSE CONTENTS**THEORY**

- UNIT 1** Matrices: Review of matrices, elementary operation on rows and columns, inverse of matrix, Normal forms, Linear dependence, Rank, Application to theory of solutions of system of linear equations, linear transformation, orthogonal, Unitary and Hermitian matrices, Characteristic equation, Eigen values and Eigen vectors, Cayley-Hamilton theorem.

- UNIT 2 Ordinary Differential Equations: Formation of differential equations, differential equations of first order and first degree (Variable separable, Homogeneous, Linear and Exact)., Linear differential equations with constant coefficients.
- UNIT 3 Ordinary Differential Equations (continued): Linear differential equations with variable coefficients, Simultaneous differential equations, Method of variation of parameters, Application to simple problems.
- UNIT 4 Probability and Statistics: Binomial, Poisson and Normal distributions and their Mean and Variance, Methods of least squares and curve fitting.
- UNIT 5 Complex Numbers: Algebra of complex numbers, Exponential function of a complex variable, Circular function of a complex variable, Hyperbolic function and inverse hyperbolic functions, Logarithmic function of complex variable, Summation of series by C+iS method.

ASSESSMENT

1. Internal Assessment for continuous evaluation, mid-term tests, tutorials, class performance, etc. (30%)
2. End semester Theory Exam (70%)

TEXT BOOKS RECOMMENDED

1. B. V. Ramana, Higher Engineering Mathematics, Tata McGraw Hill Publishing Company Ltd., New Delhi, 2006.
2. H. K. Das, Advanced Engineering Mathematics, S. Chand and Company Ltd., New-Delhi, 2009.

REFERENCE BOOKS

1. Zafar Ahsan, Differential Equation and their Applications, Prentice Hall of India Pvt. Ltd., New Delhi, 2004.
2. M. Ray, H. S. Sharma and S Chaudhary, Mathematical Statistics, Ram Prasad and Sons, Agra, 2004.
3. S Pal and S. C. Bhunia, Engineering Mathematics, Oxford University Press, New Delhi, 2015.

Revised Syllabus (as Passed in BoS Meeting held on 15/07/2022)

S.G.S.I.T.S/Syllabus/Applicable from Session 2022-2023

B.Tech. I Year (4YDC)

CH 10506: Chemistry

*Hours per week			Credits		Maximum marks				
L	T	P	Th.	P	Theory		Practical		Total Marks
3	1	2	4	1	CW	END SEM	SW	END SEM	150
					30	70	20	30	

Pre-requisite: Knowledge of basics of Chemistry studied in Class XI and XII.

Course Objectives: This course aims at imparting knowledge of the subject to the students for making them understand the role of chemistry in the field of engineering. The focus is on developing capabilities of students to use various analytical techniques, their applications in characterization of various materials used in different fields of engineering.

Course Outcomes: The course will enable student to understand:

- Basics of Organic, Inorganic and Physical Chemistry.
- Specifications, testing and treatment of water for industrial and domestic use.
- Concepts, manufacturing and applications of different types of industrially important materials and their maintenance.
- Applications of spectral and analytical techniques in qualitative and quantitative analyses.
- The students will understand the Ethics behind applying Chemical Methods in Industries and day-to-day use.

Course contents:

(i) Theory:

UNIT I- Basics of Chemistry

Organic Chemistry: Concept of Hybridization, Configuration (including R/S and E/Z nomenclature of isomers) and Conformation (Newman and Sawhorse Projection Formulae, with nomenclature)

Physical Chemistry: Rate Law, Molecularity and Order of Reaction, First Order Kinetics

Inorganic Chemistry: Theories of Chemical Bonding (VBT and MOT)

Green Chemistry : An overview

UNIT II

Water:

Source, Types of impurities and their effects, Hardness, its expression and determination, Boiler troubles and their causes, Analyses and treatment of water for industrial and domestic purposes, Alkalinity and its determination, Numerical problems. IS Specifications for Water, **Methods for disposal of waste water.**

UNIT III

(a) Lubricants

Types of lubricants and principles of lubrications, properties (test) of lubricants, greases, graphite, cooling liquids and cutting fluids and their applications. Selection of Lubricants, **Ethics code and Ethics management in the oil and lubricant industries.**

(b) Corrosion

Principle of corrosion, types of corrosion, factors affecting and methods of protection. **Ethics for corrosion prevention.**

UNIT IV

Materials Chemistry

(a) Polymers and Polymerization

Introduction, Classification, Types, mechanism, methods of polymerization. Structure-property Relationships, compounding, general applications of polymer materials of industrial importance. Concept of Biodegradable polymers, **Environmental regulations for polymer based packaging materials .**

(b) Nanomaterials

Introduction, synthesis, properties, nano-structured materials and their applications. Introduction to Smart materials and their applications.

UNIT V

Spectroscopic Techniques and Applications

Introduction to Spectroscopy, Principal, Instrumentation and Applications of UV, IR and AAS. Introduction to Chromatographic Techniques (Paper, GLC and Column), **Use of non destructive method.**

Assessment (Theory): Attendance, class test, class ass

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES I Year BE (4YDC)
HU 10651 TECHNICAL ENGLISH

HOURS PER WEEK			CREDITS			MAXIMUM MARKS				
T	P	Tu	T	P	Tu	THEORY		PRACTICAL		TOTAL MARKS
						CW	END SEM	SW	END SEM	
3	2	-	3	1	-	30	70	20	30	150

PRE-REQUISITE:

The student must possess basic knowledge of English Grammar and basic ability to read, write and speak English language

COURSE OBJECTIVES:

The objective of this course is to enable the students to recognize the relationship of effective communication skills to succeed in academic, work and social environments and to develop both written and oral communication skills to comprehend and produce clear, complete and accurate messages. This course also intends to impart business correspondence and improve English phonetics enabling the students to speak and write English correctly and with confidence.

COURSE OUTCOMES:

After successful completion of the course, *students will develop competency in the following areas:*

- CO1: Communicate effectively with good interpersonal skills; speak in public, engage the audience, make a presentation using Power-Point and tackle group discussion.
- CO2: Speak English with clarity using correct pronunciation and accent in all situations
- CO3: Comprehend a particular piece of information, absorb what is read and hone the art of condensation and effective writing skills with correct syntax and grammar in all formal situations.
- CO4: Collect, analyze, document, and report information clearly, concisely and logically. Also, apply technical information and knowledge in practical documents.
- CO5: Produce clear, complete, concise, concrete and precise messages in day to day business transactions while drafting applications and E-Mails.

COURSE CONTENT:

UNIT I: Communication Skills

- (i) Meaning & Process
- (ii) Significance of Effective communication.
- (iii) Presentation Strategies: Techniques of Oral Presentations, Defining the Purpose, Analyzing the Audience, Establishing the main idea, Selecting the Appropriate Channel and Medium. Power Point Presentation.
- (iv) Group Discussion.

UNIT II: Application of Linguistic Ability

- (a) English IPA (Types of speech sounds.)
- (b) Monophthongs
- (c) Diphthongs.
- (d) Consonants.
- (e) Phonetic Transcription

UNIT III: Advanced Writing Skills:

(i) Composition :

(a) Meaning & Significance

(b) Exercise on free essay writing on technical and current topics with particular emphasis on clear expressions.

(ii) Précis

(a) Reading and Comprehension

(b) Basic rules and practice of writing précis.

UNIT IV: Academic Writing and Technical Report Writing:

(i) Introduction to Technical and Academic Writing

(ii) Features & Principles of Technical Report

(iii) Structure & Format of Technical Report

(iv) Samples & Models for practice

UNIT V: Business Correspondence

(i) Business & Official letters

(ii) Structure & Layout of Business letters

(iii) Types of Letters: Job Application, Letters of enquiry, Quotations, Orders and Complaints.

(iv) E-Mail writing

ASSESSMENT: Through End-Sem. Theory Exam, Theory sessionals, Mid-Sem Tests, Assignments, End-Sem. Practical Exam, Practical sessional, quizzes, and presentations.

BOOKS RECOMMENDED:

1. Sharma, R.C. and Krishna Mohan, *Business Correspondence and Report Writing*, New Delhi: Tata McGraw-Hill, 2009.
2. Raman, Meenakshi and Sangeeta Sharma, *Technical Communication*, New Delhi: Oxford Univ. Press, 2011.
3. Kapoor A.N., *Business Letters for Different Occasions*, New Delhi: S. Chand & Co. Pvt.Ltd., 2012.
4. Thomson, A.J. and A.V.Martinent, *A Practical English Grammar*, New Delhi: Oxford University Press, 1986.

REFERENCES RECOMMENDED:

1. Lesikar, Petlit, and Flatley, *Lesikar's Basic Business Communication*, New Delhi: Tata McGraw-Hill, 1999.
2. Bhatia, R.C., *Business Communication*, New Delhi: Ane Books Pvt. Ltd., 2012.
3. Magan, Sangeeta, *Business Communication*, New Delhi: Biztantra Publications, 2010.

DEPARTMENT OF COMPUTER ENGINEERING
CO 10504: COMPUTER PROGRAMMING

*Hours per Week			Th. Credit	Pr. Credit	MAXIMUM MARKS				
L	T	P			TH	CW	SW	Pr.	Total
3	0	2	3	1	70	30	20	30	150

PRE-REQUISITES: NIL

COURSE OBJECTIVES: This course aims to provide exposure to problem-solving through programming. It aims to train the student to the basic concepts of the C-programming language.

COURSE OUTCOMES:

After completing the course student should be able to:

1. To formulate simple algorithms for arithmetic and logical problems.
2. To translate the algorithms to programs and test and execute the programs and correct syntax and logical errors.
3. To apply programming to solve conditional and iterative statement, function and recursion.
4. To use arrays, pointers and structures to formulate algorithms and programs.

COURSE CONTENTS:

THEORY:

- UNIT 1** Block Schematic of digital computer and it's working. Introduction to computer hardware and software, Different number systems. Flowchart and algorithm.
- UNIT 2** Structure of C programs, key words and identifiers, constants, variables, Data types, enumerated data types, Strings. Declarations of variables, scope and life of variables. Various types of operators and expressions. Programming errors and their handling.
- UNIT 3** Decision making and Branching: if-else, switch-case, Looping: While-do, for, do-while etc., nesting of loops.
- UNIT 4** Introduction to Arrays, Structures, Pointers, Files, Functions, Recursion.
- UNIT 5** Introduction to Object oriented Programming paradigm, Comparison of Procedural and Object Oriented Programming paradigm.

COURSE ASSESMENT (Th.):

1. Internal Assessment for continuous evaluation, mid-term tests, Tutorials, Quizzes, ClassPerformance, etc. (30%)
2. End semester Theory Exam (70%)

COURSE ASSESMENT (Pr.):

1. Internal Assessment for continuous evaluation (40%): Lab assignments, demonstration, Viva, file etc.
2. End semester Practical Exam (60%): Quiz/Programming test, lab journal, demo, viva etc.

TEXT BOOKS RECOMMENDED:

1. E. Balagurusamy, "Programming in ANSI C", Seventh Edition, Tata McGraw Hill, 2017.
2. Reema Thareja, "Programming in C", Second Edition, Oxford publication, 2016
3. W. Kernighan and Dennis M. Ritchie, "The C Programming Language", Pearson, 2015

REFERENCE BOOKS:

1. Matthias Felleisen, Robert Bruce Findler, Mathew Flatt, Shriram Krishnamurthi, "How to Design Programs: An Introduction to Programming and Computing", Second Edition, MIT Press, 2018.
2. M. Chandwani, A. Jain and N.S. Chaudhary, "Elements of Computer Science", Jain Publishers 1997
3. E. Balagurusamy, "Object Oriented Programming with C++", Tata McGraw Hill, 2009
4. B.S. Gottfried, "Programming with C", 3rd edition, Tata McGraw Hill, 2018

Shri G.S. Institute of Technology & Science, Indore
{{Industrial & Production Engineering Depatt.}}

B. Tech. I YEAR (4YDC)

Subject Code	Subject Name	L	T	P	Theory	Practical	Maximum Marks				
					Credit	Credit	Th	CW	SV/	Pr	Total
IP -10581	Manufacturing Practices		1	2		1.5	0	0	40	60	100
CO 1	Understanding nature of wood/timber for different wooden products along with proficiency to use different carpentry tools and perform operations related to wood working										
CO 2	Basic knowledge of casting process along with use of moulding tools for mould preparation as well as identifying casting defects and their causes during casting process										
CO 3	Basic knowledge and performing forging and welding operations using different tools/equipment along with understanding defects and their causes										
CO 4	Knowledge of fitting operation and using its tools as well as fundamentals of machining operations and conducting experiments on lathe, drilling, shaper and planner machine tool along with use of single point cutting tools and their materials										

Theory and practice:

1. Wood working : Applications of wood/timber as an engineering material, natural wood, their types, hard and soft wood. Growth of trees and their lives. Defects in wood and reasons for their occurrence, conversion of wood into timber, seasoning of timber, introduction to wood working tools for various operations like saw, chisel, jack plane, marking and measuring tools. Types of joints and their use. Manufacture of plywood and ply-boards etc. Preservation of timber, glues and adhesives in wood working, use of nails, screw etc., introduction to paints and varnishes for protection and decorative purposes
2. Foundry process : Introduction to moulding and casting processes. Moulding sand and its main constituents, moulding tools and their use, green sand moulding practice, bench and pit moulding, properties of moulding sand, use of additives in moulding sand, use of patterns and their types, pattern materials. Use of cores and chaplets. Pouring of metal and removal of castings. Casting defects and their remedies.
3. Forging process : Hot and cold working of metals, material suitable for forging operation, simple forging operation with use of tools like hammers, anvil, swage block, dies, tongs, punches, chisel, drifts etc. Forging operation like upsetting, bending and drawing out etc.
4. Welding process : Introduction to welding practice and classification (fusion and pressure welding). Types of electrodes and shielding of arc. Manual Metal Arc Welding (MMAW). AC and DC welding, tools and equipments used in electric arc welding. Types of welded joints, joints preparation and welding positions. Gas welding - Oxyacetylene welding. High and low pressure

gas welding (Constructional features of nozzles used in them). Types of flames and their use. Gas cutting. Defects in welding.

5. Machine shop : Introduction to chip removal processes, working of simple machine tools like lathe, drill, shaper, planner. Single point cutting tool and their materials and various angles. Simple lathe operations like turning, facing, taper turning, boring, threading, knurling etc. Use of taps and dies, drills and center drills and reamers on lathe. Quick return mechanism used in shaper and planner and simple operations on these machine tools.
6. Fitting practice : Need of fitting shop, common measuring and marking tools like Engineer's scale, try square, surface plate, marking gauge, V-block, center punch. Use of inside, outside and odd leg callipers, micro meters and verniers, combination set, bevel protector. Use of files of material removal, their types in respect of length and cross section, single and double cut file, filing techniques. Use of hacksaw, drills, tapes and dies.

NOTE : On an average 3 to 4 lectures/demonstrations/practice are to be delivered for each of the above process.